Geometric Algebra

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1. Algebra: The Written Form of Geometry

The invention of a written form of spoken language was one of the most important developments in human history. We refer to it as the beginning of recorded history. It allowed knowledge to be passed down from generation to generation much more rapidly and accurately than was possible by the spoken word. With hindsight we can now see it as a memory upgrade for the human mind.

The first written languages consisted of pictograms like Egyptian hieroglyphics. It took many more millennia for the Phoenicians to discover a new form capable of writing down any known language. They realized that foreign words could be sounded out by associating each discrete consonant and vowel sound with its own pictogram. This marked the birth of alphabetic languages. Any human language could now be written down. But even more importantly this was also true of words that had yet to be coined since they would still be uttered using the same discrete sounds and therefore could be written down using the alphabet. This made alphabetic languages universal.

A similar jump to universality occurred when Hindu mathematicians discovered the number zero around 900AD. For the first time any number could be written in a unique way without resorting to tallying symbols by using zero as a place holder such as 21, 201, 2001, 20001 etc. Western musical notation lead to a universal way of writing down music of virtually any genre.

A similar change occurred for writing down geometry. By about 300 BCE the Greek mathematician Euclid had developed a system for accurately constructing geometric shapes on a flat plane using only two tools, the straight edge and compass. He compiled most of what was then known about mathematics into a book called The Elements. At the time it represented the deepest knowledge that humans had about any topic. Over the next two thousand years, if anyone learned mathematics they did it by reading Euclid's book. It was one of the most printed books in history next to the Bible.

Historically, algebra and geometry developed separately with geometry being laid down by Euclid and algebra arriving well over a thousand years later. Algebra began with the addition and multiplication of numbers and developed into a highly abstract subject with many specialized branches, all with their own bewildering array of notations. When Engineers or Physicists apply mathematics to solve problems they must often master many algebras learned laboriously over many years such as: scalar algebra, the algebra of Complex numbers, the algebra of Quaternions, exterior algebra, vector algebra, linear algebra, matrix algebra, tensor algebra, Lie algebra and Clifford Algebra (to name just a few!). The last one is especially interesting. William Kingdon Clifford, a British mathematician, developed his algebra around the 1870´s and named it ***Geometric Algebra***. It was not only capable of unifying all of the algebras just mentioned into a simple and elegant notation it had a direct and intuitive connection to geometry.

In a sense Clifford discovered a simple universal written language for a vast class of geometries, not just Euclid's ruler and compass geometry. Tragically Clifford died young and his achievements were largely ignored until recently. It should be taught in elementary school along with Euclid's geometry but sadly we still subject our poor children to the standard math curriculum. Hopefully we can change that. We will make the claim that Algebra should be viewed as the written form of Geometry just as alphabetic language is a written form of the spoken word. The following is a short informal introduction to Geometric Algebra.

1.1. Euclid's Five Postulates of Geometry (Reimagined)

Euclid's book begins with some definitions of points and lines as well as stating five postulates. These are conjectures. There is no way to know if they are true in any absolute way but they can be used to deduce propositions about geometry which may lead to contradictions with other good explanations. This could mean that one or more of the postulates are wrong. As far as we know Euclidean geometry is self-consistent and is an example of an excellent explanation for how geometry works. It became the model for virtually all future mathematics.

With the benefit of hindsight we are going to restate the postulates with some modern additions which can be used to greatly extend the reach of Euclid's original postulates.

1.2. Postulate One: Vectors

A straight edge can be used to construct a directed line segment from a starting point P_1 to an end point P_2. We shall call directed line segments ***vectors*** . Vectors are usually labeled with lowercase single characters (a,b,..) with an optional arrow accent over the label if necessary. Vectors are drawn like an arrow with the arrow head at the end point as show in Figure 1.

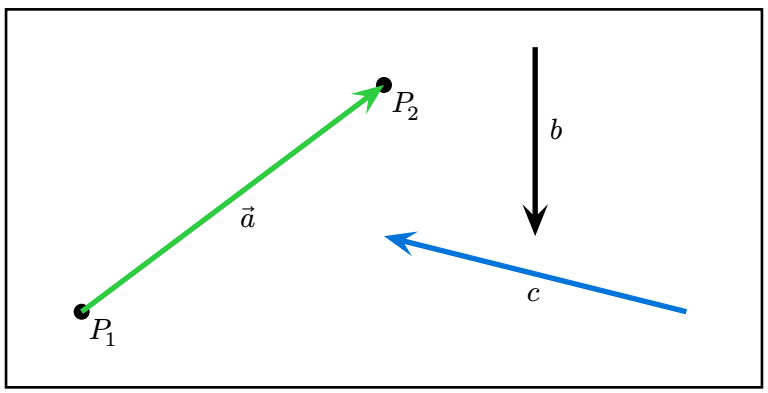

Figure 1: Vectors

1.2.1. The Principle of Equal Vectors

Two vectors can be compared by bringing together their starting points without changing their directions or lengths. Only if the vectors overlap exactly are they equal. If they have different lengths and/or directions they are not equal. Moving vectors in this way is called ***translation***. If the two vectors are equal then they can share the same label. The two vectors labeled $\vec{e}$ in Figure 2, are equal since they have the same direction and length. The same is true with the vectors labeled $\vec{f}$. You are free to move/translate a vector wherever you want as long as the length and direction remain unchanged.

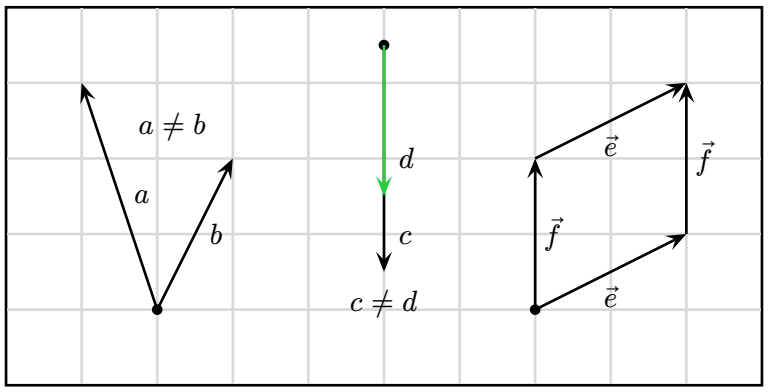

Figure 2: The Principle of Equal Vectors

1.3. Postulate Two: Lines and Points

A straight edge can be used to extend a vector indefinitely in either direction to specify a line. Note that a line and a vector are not the same kind of element since a vector has a definite length. Two arbitrary lines will intersect at a unique point except for parallel lines - see Section 1.6. Elements like lines and points will usually be labeled with uppercase letters. Some examples are shown in Figure 3. The line L is formed by extending vector $\vec{v}$ with a straight edge and the two lines L_1 and L_2 intersect at the unique point P.

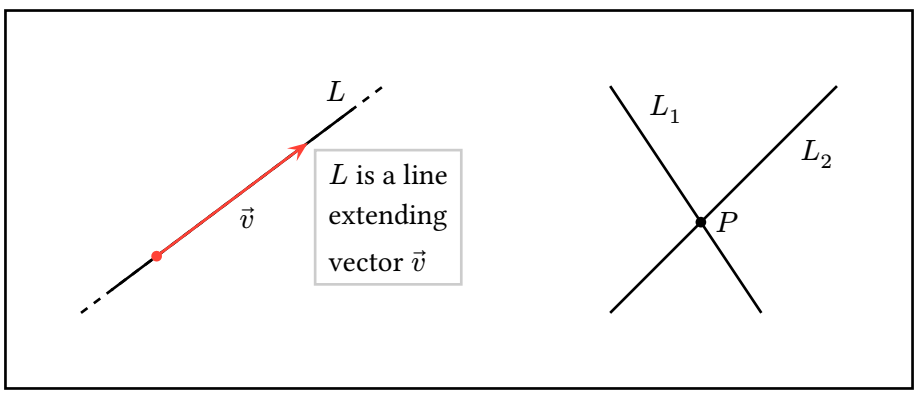

Figure 3: Lines and Points

1.4. Postulate Three: Arcs

A compass can be used to transfer the distance between two points from one place to another. Set the compass to the desired distance and move the compass needle to a location P_1 and set the marker down at a location P_2 and then sweep the marker around P_1 in the clockwise (CW) or counter clockwise (CCW) direction to a another position labeled P_3 as shown in Figure 4. All points along the arc are equally distant from the center P_1 and that distance is called the **radius** of the arc.

Use Postulate One to create a starting vector from P_1 to P_2 labeled $\vec{n}_1$ and an ending vector from P_1 to P_3 labeled $\vec{n}_2$. An arc will be defined as the product of the starting vector and the ending vector. In Figure 4 the arc labeled A is written as $A = \vec{n}_1\vec{n}_2$. Placing two labels side by side implies multiplication. An explicit symbol for multiplication is usually not used. The order of the two vectors matters. The arc goes from $\vec{n}_1$ to $\vec{n}_2$ in the CCW orientation as indicated by the blue arrow head.

Vectors labeled with n or $\vec{n}$ will have a length of one unit. The unit of length can be freely chosen for convenience. Such vectors are usually called **unit vectors**.

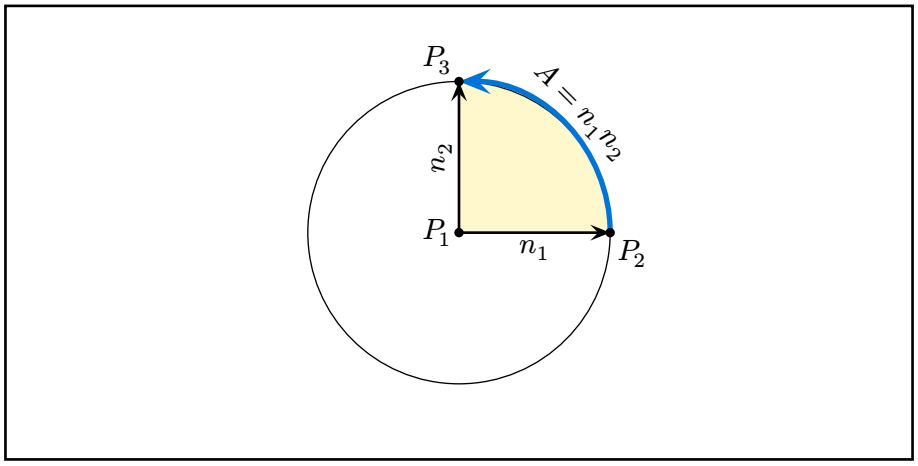

Figure 4: The Compass and Arc

We know how to multiply numbers but you might be wondering what we mean by "multiplying vectors". This will be described in detail later but for now the product of two vectors is simply how we write down a circular arc in Euclidean geometry.

The arc is the circular curve swept out by the compass on the perimeter of the circle (the blue arrow labeled A). It can be thought of as a curved vector or as a sector of the circle like a pizza slice (the yellow region in Figure 4). Arcs are directed in the clockwise (CW) or counterclockwise (CCW) orientations. They have a starting vector and an ending vector. The length of the arc is proportional to the angle between the vectors. The area of the sector is also proportional to the angle. Arcs will usually be labeled with uppercase characters and the angle of an arc with a lowercase t. For example the angle of the arc A would be labeled t_A.

Note that the angle of the arc is a number: $t_A = 90°$ but the arc is the product of two vectors and contains more information than just the angle. For this reason, arcs will be labeled differently than angles.

We call the product of two vectors the ***Geometric Product***. It is one of the most important ideas in all of mathematics. One can think of the product of two numbers as a special case of the geometric product.

1.5. Postulate Four: The Principle of Equal Arcs

An arc can be moved around the circle without changing its value, just as a vector can be translated along a line without changing its value. In Figure 5 we have two equal 90° arcs. The arc $n_1 n_2$ is rotated by 180° CCW by rotating n_1 to n_3 and n_2 to n_4. We could have rotated it by any angle we want but 180° was chosen to make it clear that both arcs are equal. Even though all four vectors are different, $n_1 n_2 = n_3 n_4$ must hold for the arcs to be equal. The geometric product must be defined in such a way as to guarantee the equal arcs principle.

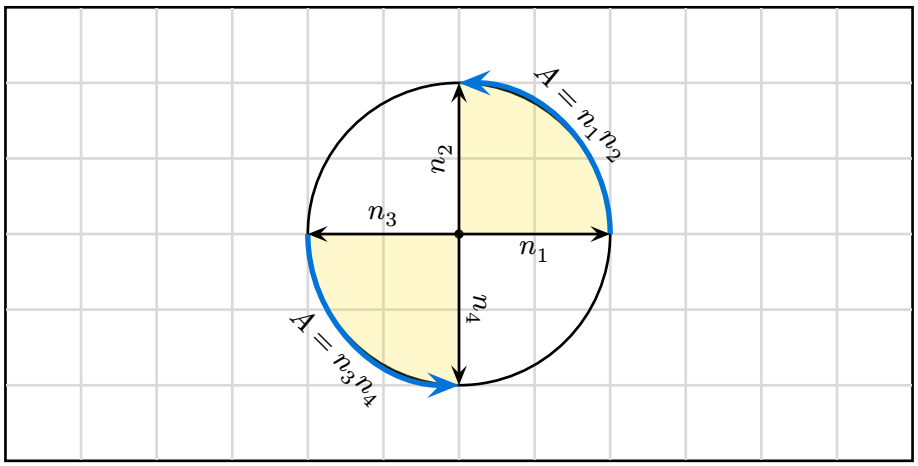

Figure 5: The Principle of Equal Arcs

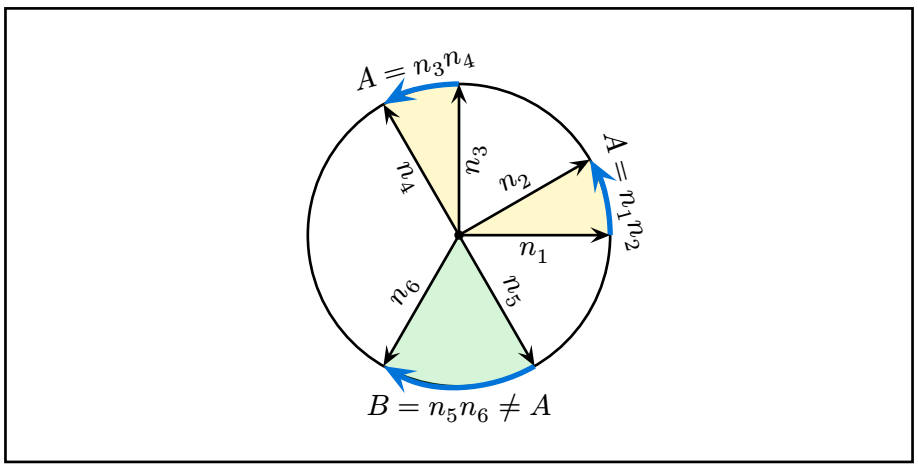

Figure 6: Example Arcs

An arc contains three pieces of information, its radius, its plane and its angle. In Euclid's plane geometry there is only one plane but in a three dimensional world there are an infinite number of planes. The two vectors of the arc will always specify a unique plane.

1.6. Postulate Five: Parallel Lines

Given a line L_1 and a point Q, not on that line, Postulate Five demands that there exists a unique line L_2 passing through point Q that does not intersect L_1 as shown in Figure 7.

This postulate became quite controversial among mathematicians throughout the following millennia. Many thought that it was redundant as it seemed that a parallel line could be constructed using the previous postulates. It turned out that Euclid was quite right to insist on the parallel lines postulate. It is crucial to the development of a Euclidean geometric algebra as we shall see.

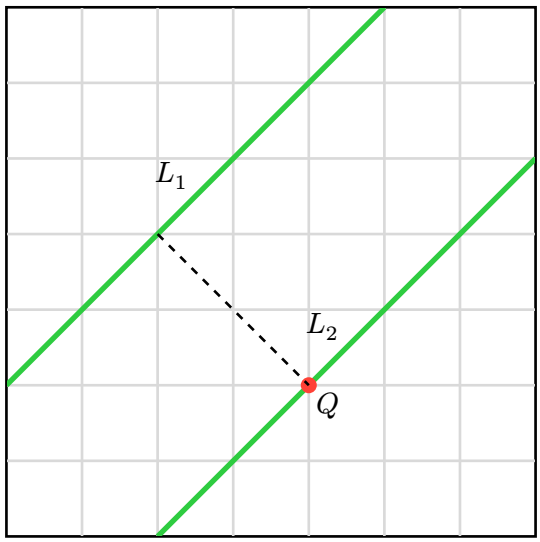

Figure 7: Parallel Lines

1.7. A Universal Written Language for Geometry

Our goal is to use these postulates to find a universal written notation for geometry using vectors and arcs as our building blocks.

If we draw any one of Euclid's constructions on a page, we are free to slide the page to a new position and rotate the page around any point. When we do this, all the sizes and shapes of the elements in the construction remain the same. There should be no preferred point of view for the space. These properties must be reflected in the language of geometric algebra. I use the word reflected purposefully - more on that later.

[Teachers note: we assume that the student has experience with using a ruler and compass to draw many of the constructions in Euclid's book. There are many great videos of Euclidean constructions online.]

2. Vectors

2.1. Two Tools: Addition and Multiplication

Euclid realized that with only two tools, the straight edge and the compass, he could construct the rest of the geometric objects in plane geometry (like circles, arcs, line segments, lines, triangles, rectangles, etc). In our new language of geometry called algebra we will follow Euclid and use only two tools called addition and multiplication. We begin with the vectors we met in Postulate One.

Vectors are everywhere in the physical world. The velocity of a car has a magnitude and direction which can be nicely modelled with a vector. The same is true for many other physical quantities like forces, displacements, electric and magnetic fields etc. Vectors will also be our building bocks for geometry so we need to learn how to add them and multiply them.

2.2. Addition

The sum of two vectors labeled a and b is written as $a + b$. To construct it we translate b so that it starts where a ends. Recall that translating a vector will not change its length or direction. Euclid provides a construction for translating line segments using a compass and straight edge. The resulting sum is a new vector from the start of a to the end of b, which we'll label c. The sum is written as $c = a + b$. The sum of two vectors is always another vector. It's easy to see that the order of the two vectors does not matter as shown in Figure 8, $c = a + b = b + a$.

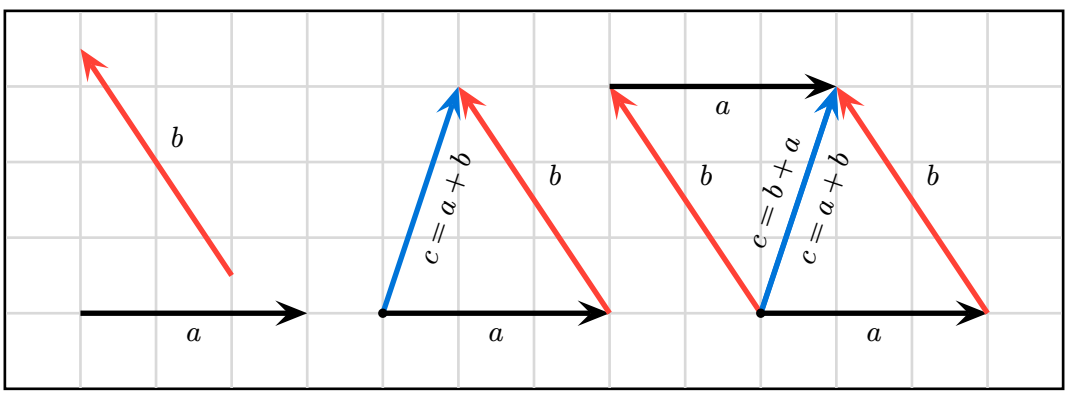

Figure 8: Adding vectors

When adding 3 or more vectors they are all translated to be "head to tail" and the sum is again another vector from the start of the first vector to the end of the last vector as shown in Figure 9.

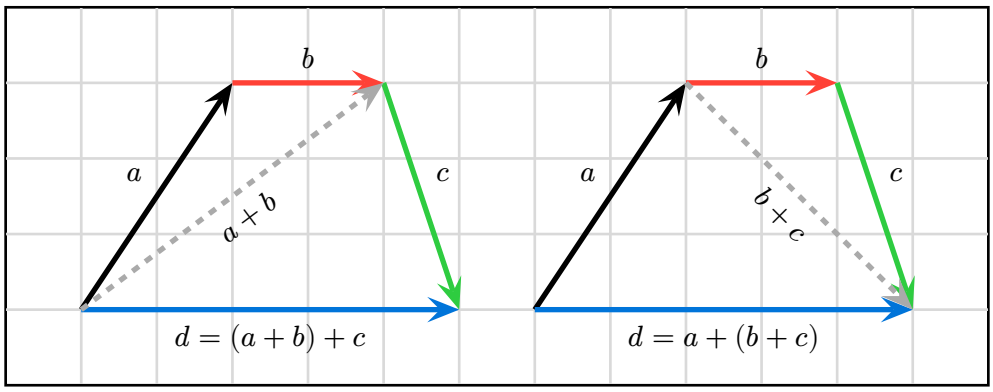

Figure 9: Adding three or more vectors

We are free to add a and b first and then add c to that. This is written as $d = (a + b) + c$ where the expression inside the parentheses is evaluated first. We can also choose to evaluate $b + c$ first and add it to a, written as $d = a + (b + c)$. The result is the same in both cases.

$$\begin{aligned} d &= (a + b) + c \\ &= a + (b + c) \\ &= a + b + c \end{aligned}$$

The sum of two arbitrary vectors can always be constructed using Euclid's straight edge and compass. Give it a try!

2.3. Scaling

Multiplying a vector by a number will scale the vector by that amount. Scaling a vector by one leaves its length unchanged, scaling it by 2 will double its length etc. The order of multiplication makes no difference when scaling. These numbers are often called ***scalars***.

$$\begin{aligned} 1a &= a1 = a \\ 2a &= a2 = a + a \end{aligned}$$

When adding scaled versions of the same vector we can just add the scalars.

$$2a + 3a = (2 + 3)a = 5a$$

A ***unit vector*** is a vector with a length of one unit and is often labeled with a lowercase n, like n_1 or with a lowercase e such as e_1, e_2, etc. Scalars will usually be labeled with lower case characters in the middle of the alphabet like r, s, and t. If the situation is ambiguous we may put an arrow accent over the vector label: $\vec{n}_1$. Examples are shown in Figure 10.

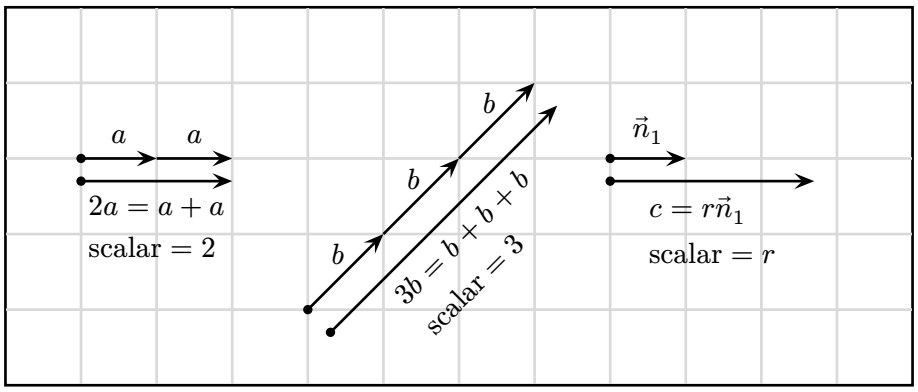

Figure 10: Scaling vectors

Vectors can always be written in two forms, as a sum of two or more other vectors or as a scaled unit vector: $c = r\vec{n} = a + b$. The unit vector has the same direction as vector c as shown in Figure 11.

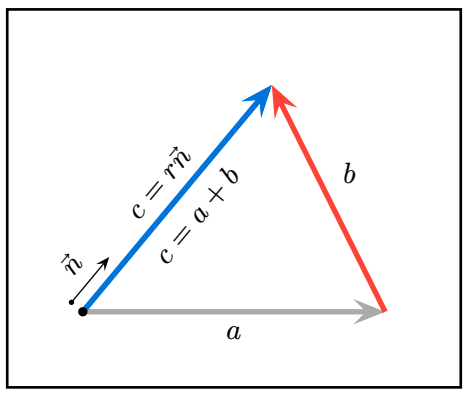

Figure 11: Vector forms

2.4. Multiplying Scalars

Scaling a vector by 2 and then scaling that by 3 is equivalent to scaling the original vector by $2 \times 3 = 6$.

$$3(2a) = 2a + 2a + 2a = 6a = (3 \times 2)a$$
$$2(3a) = 3a + 3a = 6a = (2 \times 3)a$$

The product of two scalars will always be a scalar and the order of the scalars doesn't matter (note: this is ***not*** true for multiplying vectors). We are free to scale any geometric object. For example scaling an arc labeled A by 3 is written as $3A = A3$.

Arbitrary geometric elements like points, lines and arcs will be labeled with upper case characters. When geometric elements are multiplied together, if any of the elements are scalars then they can be moved to any position in the product. It is often useful to move all the scalars to the left side of the product and replace them with one scalar (their product). For example:

$$
\begin{aligned}
C &= 3A4B \\
&= 3(A4)B \text{ ; select the middle two elements in the product} \\
&= 3(4A)B \text{ ; using } A4 = 4A \\
&= (3 \times 4)AB \text{ ; multiply the first two scalars in the product} \\
C &= 12AB
\end{aligned}
$$

The freedom to move scalars wherever we like in a product is an extremely important ***geometric*** property which is seldom emphasized.

2.5. Point Reflections and Negative Scalars

A vector can point in any direction. It is often very useful to reverse the direction of a vector by swapping the head and tail. It can be constructed easily by extending the vector into a line with a straight edge (Postulate Two) and then using a compass to transfer the length of the vector to the opposite side of the start point (Postulate Three). The reflected vector has the same length as the original and lies along the same line. This is called a point reflection. It reflects the vector thru its starting point. It is written by placing a tick symbol, $-$, (usually called a minus sign or negative sign) to the left of the label. See Figure 12 left.

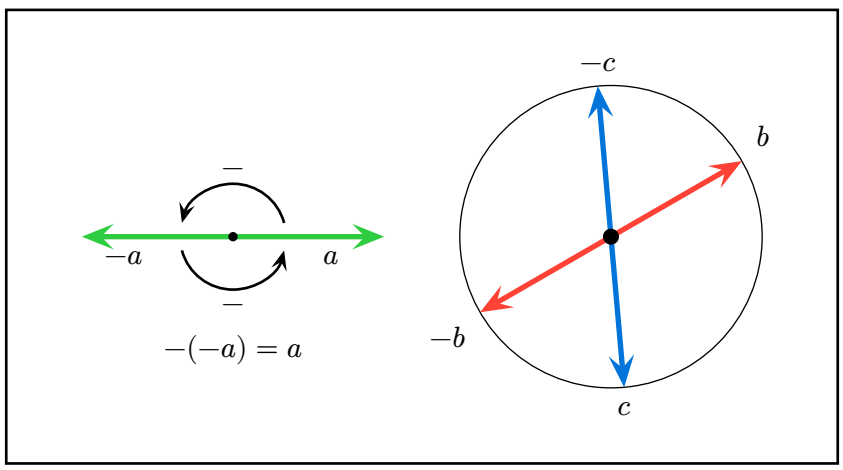

Figure 12: The point reflection of vectors

You should think of the tick symbol, $-$, as an operation on the vector. It means "reflect the vector thru its starting point". Two consecutive reflections cancel since the vector will return to its original orientation.

$$-(-a) = (--)a = (1)a = a$$

Thus we will ***define***: $(--) = 1$, the product of two "ticks" equals 1. You may recall your math teacher telling you that "two negatives is a positive". Well now you know *why* this is true. Two consecutive point reflections leaves the element unchanged. The point reflection operation behaves like a scalar and thus obeys the same rules as a scalar. The reason for this will be made clearer in the next chapter. For example we are free to move them around in product expressions just like scalars as shown below.

$$
\begin{aligned}
(-a)(-b)c(-d) &= (---)abcd \text{ ; move all ticks to the left like scalars} \\
&= (--)(-abcd) \text{ ; the first two ticks cancel} \\
&= (1)(-abcd) \\
(-a)(-b)c(-d) &= -abcd
\end{aligned}
$$

This implies that the product of an even number of ticks will be 1 and the product of an odd number ticks will be one tick or -1.

For example, the product of four consecutive ticks is 1.

$$
\begin{aligned}
---- &= (--)(--) \text{ ; product of an even number of ticks} \\
&= (1)(1) \\
---- &= 1
\end{aligned}
$$

The product of five consecutive ticks is -1.

$$
\begin{aligned}
----- &= -(--)(--) \text{ ; product of an odd number of ticks} \\
&= -(1)(1) \\
----- &= -1
\end{aligned}
$$

A point refection times a scalar can be thought of as a composite scalar. For example $-2a$ (negative two times the vector a) will first scale a by 2 and then do a point reflection. The order doesn't matter since doing the point reflection first and then scaling by 2 results in the same vector so $-2a = 2(-a)$.

The idea of a negative number seemed very strange to our ancestors. They thought numbers were about counting things so whole numbers made sense but negative numbers where hard to swallow. It took mathematicians well over a thousand years after Euclid to come to grips with them!

However from a geometric point of view, it is perfectly reasonable to take two steps to the right, reverse direction and take 3 steps to the left for example. You end up one step to the left of your starting position. We could write this as, $2 \text{ right} + 3 \text{ left} = 1 \text{ left}$. Since right and left are just arbitrary directions we can model them with vectors and "reversing direction" can be accomplished with a minus sign in front of the vector. We can write the previous example as, $2\vec{a} + 3(-\vec{a}) = -\vec{a}$, where $\vec{a}$ is a vector pointing to the right with a length of one "step" and $-\vec{a}$ points to the left by one step.

2.6. Zero

If we add vectors that return to the starting point of the first vector, the resulting vector will have no length as shown in Figure 13. We will define a new scalar called zero written as 0 that will scale a vector to have no length or size. The vector $0\vec{a}$ starts and ends at the same point.

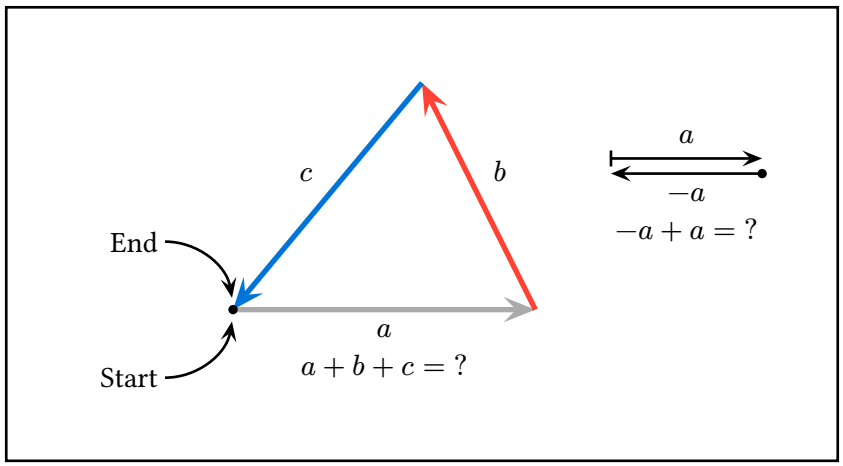

Figure 13: Zero vectors

We can extend this idea so that scaling any geometric element by zero shrinks it out of existence. We write this as $0A = 0$ for short. We will also define the sum of zero and any element as leaving the element unchanged.

$$-A + A = 0A = 0$$
$$A + 0 = A$$

Some examples:

$$-3 + 3 = 0$$
$$-2a + 2a = 0$$
$$5A + -5A = 0$$

The discovery of the number zero was one of the most important ideas in all of mathematics. It lead to the universal way we write down arbitrary numbers by using the numeral zero as a place holder as in 201 for example. This was not possible using the Roman number system. It took well over a thousand years from Euclid's time to discover these ideas.

On the other hand, adding a vector to its point reflection leads to a vector of no length. Since this can happen we must have a way of writing it down. Hence the number zero. It arises naturally when we take a geometric point of view. This is why we modified Euclid's first postulate to distinguish between the start and end of a line segment. It seems like a trivial distinction but turns out to be enormously important.

2.7. Solving Equations

The expressions on both sides of an equal sign can be changed as long as they are kept in balance. Any operation done to one side must also be done to the other in order to maintain equality. This idea goes back to the ninth century. A Persian mathematician named Al-Khwarizmi wrote a book called "Kitab al-Jabr". We get the word "algebra" from "al-Jabr". He introduced the idea of balancing equations.

For example in an equation like $a + b = c$ if we want to rearrange the expression so that the label b is isolated on the left of the equal sign, we need to get rid of the a. We can do this by adding $-a$ to both sides, keeping the equation in balance as shown below.

$$
\begin{aligned}
a + b &= c \\
(-a) + a + b &= (-a) + c \text{ ; add } (-a) \text{ to both sides} \\
(-a + a) + b &= -a + c \text{ ; use } (-a + a) = 0 \\
0 + b &= -a + c \text{ ; use } 0 + b = b \\
b &= -a + c
\end{aligned}
$$

The result is that we can move a label in a sum to the other side of the equal sign (in either direction) but we have to include a minus to keep the equation balanced.

$$
\begin{aligned}
a + b &= c \text{ ; move } a \text{ to the right side and change its sign} \\
b &= -a + c \text{ ; move } c \text{ to the left side} \\
-c + b &= -a
\end{aligned}
$$

For example if we know that $a = 2$ and $c = 5$, we can find the value b:

$$
\begin{aligned}
a &= 2 \\
c &= 5 \\
a + b &= c \text{ ; move } a \text{ to the right side to isolate } b \\
b &= -a + c \text{ ; use the values } a = 2 \text{ and } c = 5 \\
&= -2 + 5 \\
b &= 3
\end{aligned}
$$

We can use the rules of addition and multiplication to manipulate equations but we have to be careful to use them properly. Later we will see that using algebra to manipulate symbols corresponds to manipulating the geometric elements they represent.

To move scalars from one side to the other involves multiplying both sides by a fraction that cancels the scalar. We assume the reader is familiar with adding and multiplying fractions. If not then ask your AI for some help. For example, to isolate the vector b on the left we need to get rid of the scalar 3:

$$
\begin{aligned}
3b &= c \\
\left(\frac{1}{3}\right)3b &= \left(\frac{1}{3}\right)c \text{ ; multiply both sides by } \left(\frac{1}{3}\right) \\
\left(\frac{3}{3}\right)b &= \frac{c}{3} \text{ ; use } \left(\frac{3}{3}\right) = 1 \\
(1)b &= \frac{c}{3} \text{ ; use } (1)b = b \\
b &= \frac{c}{3}
\end{aligned}
$$

For example if we know that $c = 6$ we can find the value of b:

$$
\begin{aligned}
3b &= c \text{ ; isolate b on the right side by dividing both sides by 3} \\
b &= \frac{c}{3} \text{ ; use } c = 6 \\
&= \frac{6}{3} \\
b &= 2
\end{aligned}
$$

The ***inverse*** of a scalar r is $\frac{1}{r}$ because their product will be 1. Thus moving a scalar to the other side is done by dividing the other side by the scalar.

$$
\begin{aligned}
rA &= B \\
A &= \frac{B}{r}
\end{aligned}
$$

2.8. Basis Vectors

In order to do calculation involving vectors we need some way of expressing the value of a vector in a unique way. This can be done by expressing any vector as a sum of two conveniently chosen reference vectors often called ***basis vectors*** which are usually written as e_1 and e_2. It is very convenient to choose the basis vectors to be mutually perpendicular unit vectors as it greatly simplifies adding and multiplying them. The basis vector e_1 is often drawn as a horizontal unit vector and e_2 is often drawn vertically. This is just a convention, they don't have to be drawn that way. All other vectors can then be uniquely written down in terms of the basis vectors by scaling them by the appropriate amounts and adding them.

For example we can write down a triangle in terms of our basis vectors by scaling e_1 and e_2 and adding them together as shown in Figure 14.

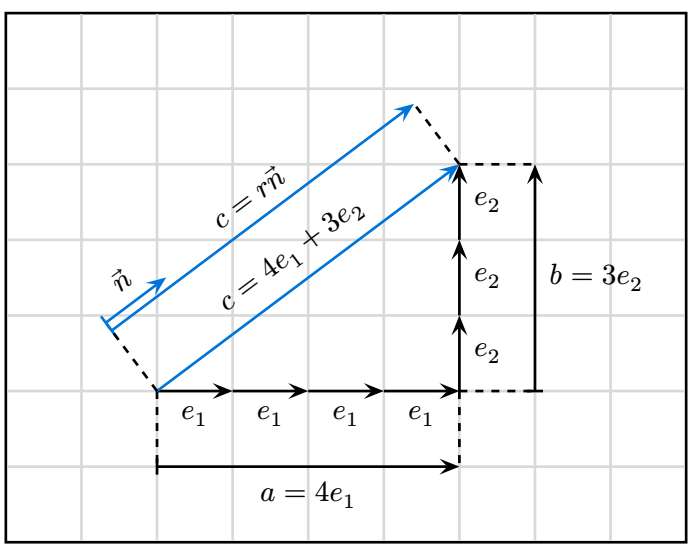

Figure 14: Basis vectors

The length of a vector is usually written as $|\vec{c}|$. Some authors prefer to write it this way: $\|\vec{c}\|$. This converts a vector into a scalar. The length of a unit vector is always 1 so $|e_1| = 1$ and $|\vec{n}| = 1$. The length of a scalar is the absolute value of the scalar so $|-3| = 3$. Finally the length of a product is the product of the lengths:

$$
\begin{aligned}
|2e_1| &= |2||e_1| \text{ ; use } |e_1| = 1 \\
&= 2(1) = 2 \\
|3e_2| &= 3 \\
|c| &= |r\vec{n}| \\
&= |r||\vec{n}| \text{ ; use } |\vec{n}| = 1 \\
&= |r|(1) \\
|c| &= |r| \\
|abc| &= |a||b||c|
\end{aligned}
$$

What is the actual value for the length of the diagonal vector c in Figure 14? Hint, it's not 7.

Unfortunately we can't answer that question yet. We will need to learn about arcs first. The next chapter will solve this problem. The earliest references to it go back to the ancient Sumerians over 5000 years ago. Euclid provided a construction showing how to get the length of a diagonal in terms of the lengths of the two shorter sides but it was quite complicated, involving many clever applications of the straight edge and the compass. This is why we need to know about arcs before we can solve this problem. Fortunately, the power of geometric algebra will make this job much easier.

See Figure 15 for another example of using basis vectors to find the sum of vectors.

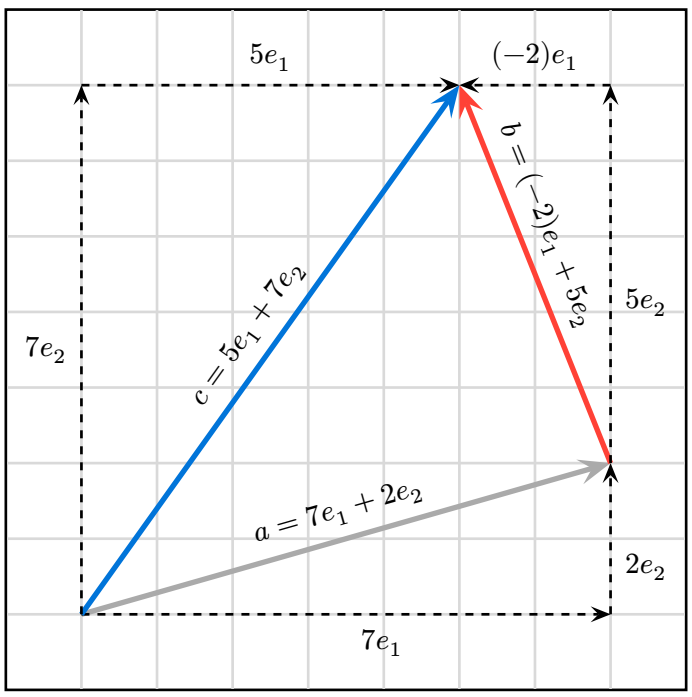

Figure 15: Adding with basis vectors

$$
\begin{aligned}
a &= 7e_1 + 2e_2 \text{ ; move 7 units right and 2 up} \\
b &= 5e_2 + -2e_1 \text{ ; move 5 up and 2 left} \\
c &= a + b \text{ ; replace } a \text{ with } (7e_1 + 2e_2) \\
&= (7e_1 + 2e_2) + b \text{ ; replace } b \text{ with } (5e_2 + -2e_1) \\
&= (7e_1 + 2e_2) + (5e_2 + -2e_1) \text{ ; group like unit vectors} \\
&= (7e_1 + -2e_1) + (2e_2 + 5e_2) \text{ ; add scalars for like unit vectors} \\
&= (7 - 2)e_1 + (2 + 5)e_2 \\
c &= 5e_1 + 7e_2 \text{ ; the result is 5 right and 7 up}
\end{aligned}
$$

Note that the parentheses in the equations are just added to show related terms. Because we are adding vectors, we are free to rearrange the terms as we like. In the last few lines we add multiples of the same unit vectors together: $7e_1 + -2e_1 = 5e_1$ and $2e_2 + 5e_2 = 7e_2$. As you become more accustomed to applying the rules of addition and multiplication you can skip most of the steps above and just write down the answer by inspection. If you get confused remember to draw yourself a vector diagram with labels as a guide.

[Teacher's note: this would be a good place to present some exercises on vector addition and scaling]

2.9. Products and Sums: Distribution

In Figure 16, the triangle formed by vectors a, b and c is scaled by 2. All the vector lengths are doubled.

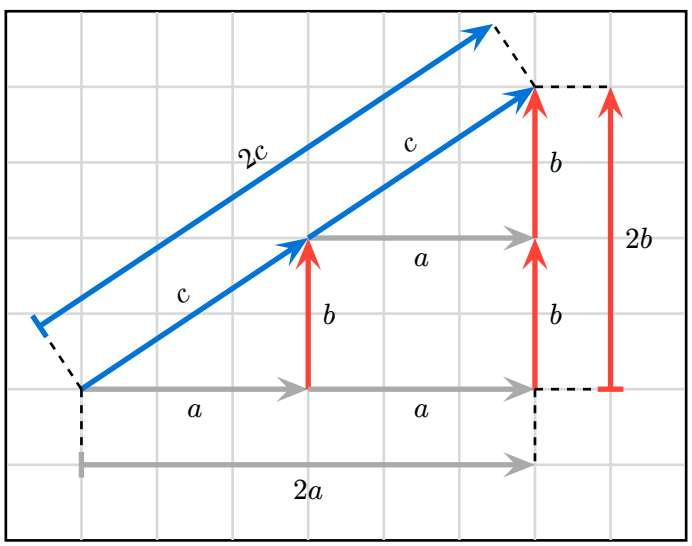

Figure 16: Distribution

$$
\begin{aligned}
c &= a + b \\
2c &= 2(a + b) \text{ ; replace } c \text{ with } (a + b) \\
2c &= 2a + 2b \text{ ; from the diagram} \\
2(a + b) &= 2a + 2b
\end{aligned}
$$

We say that the scale factor 2 is ***distributed*** across the sum $(a + b)$. Each vector in the sum is multiplied by 2. This pattern can be extended to any number of geometric elements as follows.

$$A(B_1 + B_2 + B_3 + ...) = AB_1 + AB_2 + AB_3 + ...$$

Where the element A is distributed across all elements in the sum on the left hand side, resulting in a **sum of products** on the right hand side, where element A "left multiplies" each of the B elements. The order of the multiplication is important, $AB \neq BA$ in general, as we saw with arcs. The A's and B's can be any type of geometric element including scalars. A sum of elements can be "right multiplied" as well:

$$(B_1 + B_2 + B_3 + ...)C = B_1C + B_2C + B_3C + ...$$

2.10. Products and Sums: Factoring

Factoring is the inverse of distribution and pulls a common element out of a sum of products. This is not really a new operation, it simply "un-distributes" a distribution. For example:

$$3a + 3b = 3(a + b) \text{ The common factor 3 is moved out of the sum.}$$

More generally:

$$AB_1 + AB_2 + AB_3 = A(B_1 + B_2 + B_3)$$
$$B_1C + B_2C + B_3C = (B_1 + B_2 + B_3)C$$

2.11. Product of Sums: Sum of Products

We can use the distribution rule to convert a product of sums into a sum of products. This is an extremely useful operation that often results in the ability to simplify expressions even through at first glance it appears to complicate things. We will use this a lot!

As an example consider a triangle $c = a + b$, we can get a product of sums by multiplying the vector c by itself. This is often written in *power* or *exponent* notation as a convenient shorthand:

$$cc = c^2$$
$$ccc = c^3$$
$$cccc = c^4$$
$$\cdots$$

The expression c^2 is usually pronounced "c squared". Thus $c^2 = cc = (a + b)(a + b)$. This is a product of sums. We will now use our distribution rules to convert this into a sum of products.

$$\begin{aligned}
c^2 &= cc \text{ ; replace the first } c \text{ with its value} \\
&= (a + b)c \text{ ; distribute the } c \text{ across the sum} \\
&= ac + bc \text{ ; replace the first } c \text{ with its value} \\
&= a(a + b) + bc \text{ ; distribute the } a \text{ across the sum} \\
&= aa + ab + bc \text{ ; replace the } c \text{ with its value} \\
&= aa + ab + b(a + b) \text{ ; distribute the } b \text{ across the sum} \\
&= aa + ab + ba + bb \text{ ; this is a sum of products} \\
\underbrace{(a + b)(a + b)}_{\text{product of sums}} &= \underbrace{aa + ab + ba + bb}_{\text{sum of products}}
\end{aligned}$$

In this example the result is a sum of vector products which are arcs according to postulate three. So we end up with the sum of four arcs. In the next chapter we will see that this simplifies to a single scalar related to the length of vector c! These types of manipulations are extremely important so it's good to practice using the rules in this way.

2.12. The Multiplication of Geometric Elements

We showed earlier that changing the order of the addition of elements doesn't change the result. This is also true of the multiplication of scalars but is not the case when multiplying arbitrary vectors. As explained in postulate 3 and 4, the order of the product of the unit vectors in an arc matters. However, just as we saw for addition, we can multiply three or more elements in any sequence we like as long as we don't swap adjacent elements. Mathematicians would say that both addition and multiplication are ***associative***. This is a very powerful property of the geometric product. We will employ it often.

$$\begin{aligned} d &= (a+b)+c & D &= (AB)C \\ &= a+(b+c) & &= A(BC) \\ &= a+b+c & &= ABC \end{aligned}$$

2.13. Three Dimensions

There is no need to stop at only two basis vectors e_1 and e_2. For three dimensional space we can simply add a new basis vector e_3 that we can scale to carry information about a third direction and add it to any other vector as shown in Figure 17. The vector e_3 is perpendicular to both e_1 and e_2. When we add $b = e_3$ to the red vector $a = e_1 + e_2$ we translate it to the the end of vector a. The translated vector b is shown in green. The result is the vector $c = e_1 + e_2 + e_3$ shown in blue.

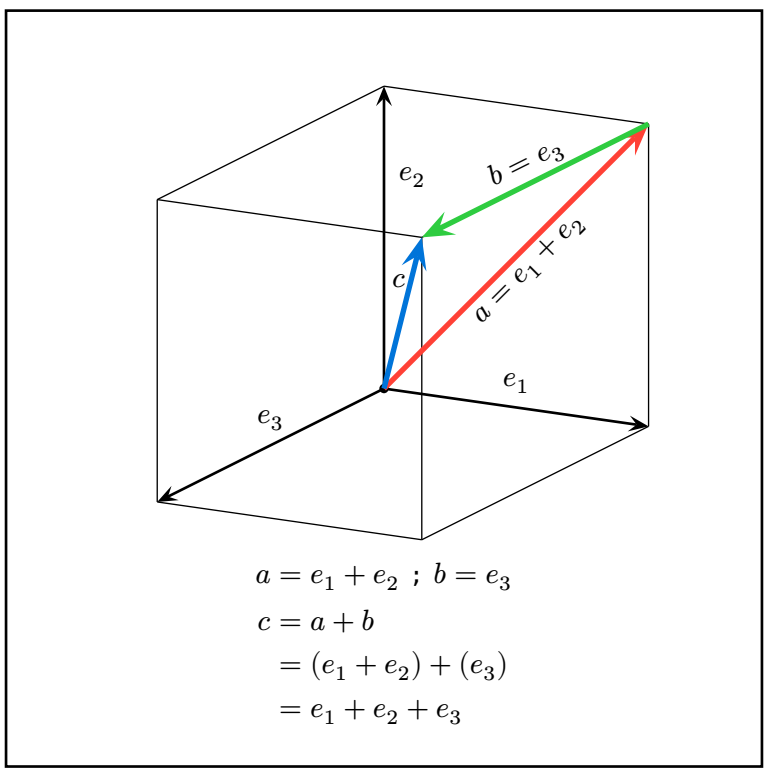

Figure 17: Three basis vectors

We are free to add as many basis vectors as we need to solve the problem we are working on. No new rules are needed since the basis vectors all behave according to the same rules. This greatly expands the range of problems we can solve far beyond Euclid's simple plane geometry.

2.14. Summary: Rules of Addition and Multiplication

Euclid used two tools for geometry, the straight edge and compass. The written form of geometry will also use two operations, addition and multiplication. The rules for addition and multiplication are summarized below. Uppercase labels refer to arbitrary geometric elements. The lowercase r refers to scalars.

- Addition

$$\begin{aligned} A+B &= B+A \\ (A+B)+C &= A+(B+C) \\ -A+A &= 0 \\ A+0 &= A \end{aligned}$$

- Multiplication

$$\begin{aligned} AB &\neq BA \\ (AB)C &= A(BC) \\ rA &= Ar \\ 0A &= 0 \\ 1A &= A \\ -(-A) &= A \end{aligned}$$

- Distribution

$$\begin{aligned} A(B_1+B_2) &= AB_1+AB_2 \\ (B_1+B_2)C &= B_1C+B_2C \end{aligned}$$

- Contraction: One Rule to Rule Them All

$$\vec{a}^2 = \vec{a}\vec{a} = r$$

The last rule is often called the ***Contraction*** rule. It will be discussed in depth in the next chapter. It is included here for completeness. It simply asserts that the geometric product of a vector with itself is always a scalar. This is one of Clifford's greatest contributions to mathematics as we shall see.

The rules of geometric algebra are simple and easy to use but have a tremendous hidden power to solve problems. The reader is encouraged to practice using them so that they become second nature. The payoff is well worth it.

[Teachers note: the sections above should be sprinkled with many more examples and exercises.]

These rules apply to vectors composed of any number of basis vectors. Thus the rules are dimension independent. They are also independent of the location, length and direction of the chosen basis vectors. When Einstein was developing his general theory of gravity he was guided by the principle that the laws of physics should be the same for all observers regardless of their point of view or motion. We are also guided by a similar principle, the laws (or rules) of geometry should be the same for all observers regardless of their point of view.

The rules are effectively the notation for a new language, the written form of geometry called Algebra. They embody Euclid's postulates and rules for building constructions using the straight edge and compass but have a much greater reach than Euclid's plane geometry.

Like any new language it takes a lot of practice to become fluent. Fortunately this language is much simpler than any written human language which any child can learn.

3. Arcs

3.1. Angles

The most common way of specifying the angle of an arc is in units called ***degrees*** which goes back to the ancient Babylonians. They divided the circle into 360 equal angles. Why divide the circle into 360 parts? The reason was that there are many ways to divide the number 360 into a whole number parts. This was very handy in a time with no calculators!

$$\begin{aligned}
360^\circ &= 2 \times 180^\circ \\
&= 3 \times 120^\circ \\
&= 4 \times 90^\circ \\
&= 5 \times 72^\circ \\
&= 6 \times 60^\circ \\
&= 8 \times 45^\circ \\
&= 9 \times 40^\circ \\
&= 10 \times 36^\circ \\
&= 12 \times 30^\circ \\
&= 15 \times 24^\circ \\
&= 24 \times 15^\circ \\
&= 30 \times 12^\circ \\
&= \ldots \\
&= 360 \times 1^\circ
\end{aligned}$$

Angles have two possible orientations, Clockwise (CW) and Counter Clockwise (CCW). By convention positive angles are usually CCW and range from 0° to 360°. Negative angles are usually CW. Thus an arc from e_2 to e_1 has an angle of 90° CW or $-90°$ for short.

The intersection of two arbitrary lines will produce two pairs of equal arcs as show in Figure 18.

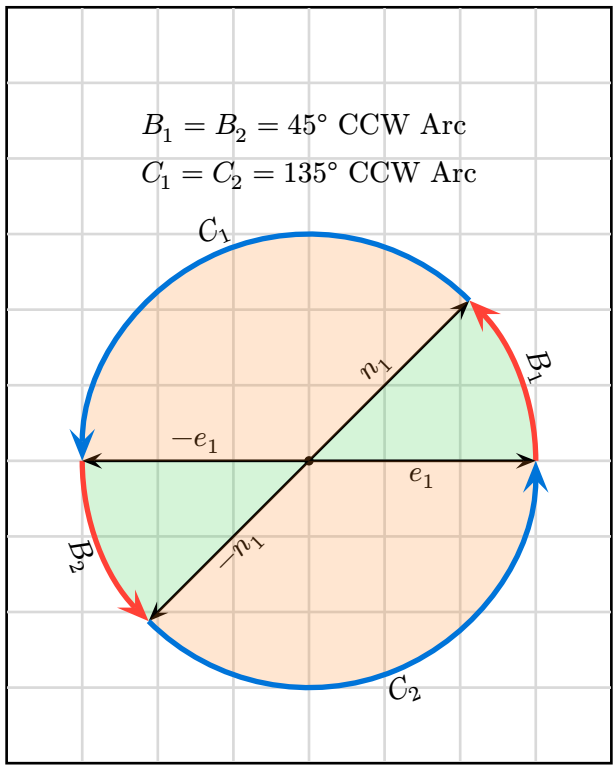

Figure 18: Opposite arcs are equal

The arc B_1 sweeps from e_1 to n_1 and so according to postulate three, we can write this as $B_1 = e_1 n_1$. Similarly, B_2 sweeps from $-e_1$ to $-n_1$ and is written as $B_2 = -e_1(-n_1)$. The two minus signs will cancel showing that $B_2 = B_1$:

$$\begin{aligned}
B_2 &= -e_1(-n_1) \text{ ; move minus signs to the left} \\
&= (--)e_1 n_1 \text{ ; use } (--) = 1 \\
&= (1)e_1 n_1 \\
&= e_1 n_1 \\
B_2 &= B_1
\end{aligned}$$

Similarly we can show that $C_1 = C_2 = -n_1 e_1$. This shows that the arcs on opposite sides of intersecting lines are equal.

3.2. Ninety Degree Arcs

Euclid drew attention to the special significance of the 90° arc in his fourth postulate. He asserted that all 90° angles are equal meaning that if two perpendicular lines intersect then all four angles between the lines are equal to 90° as shown in Figure 19. Our basis vectors, e_1 and e_2 are perpendicular. We have updated this postulate to include the orientation of the arcs as well: a CW arc is not the same as a CCW arc. All the arcs labeled A are equal since they are just rotated versions of the same arc.

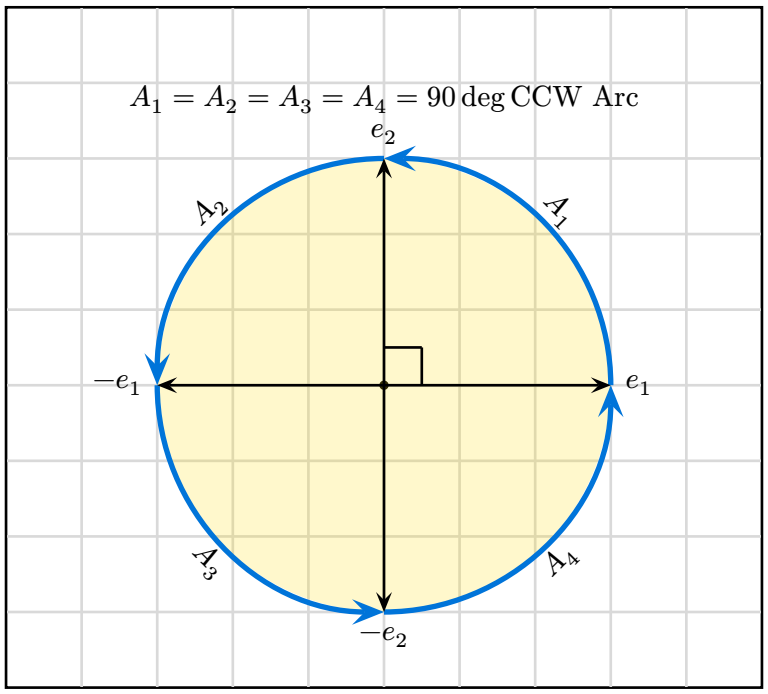

Figure 19: All 90° arcs are equivalent

Referring to Figure 19, the opposite arcs are equal as shown previously thus $A_1 = A_3$ and $A_2 = A_4$. Arc A_1 sweeps from vector e_1 to vector e_2. Thus we write this arc as $A_1 = e_1 e_2$. It sweeps 90° CCW. The arc just below it A_4 starts with vector $-e_2$ and ends with e_1 thus $A_4 = -e_2 e_1$. It also sweeps 90° CCW. These arcs must be the same as demanded by postulate four. This leads to the very important relationship between our *perpendicular* basis vectors:

$$\begin{aligned} A_1 &= A_4 \\ e_1 e_2 &= -e_2 e_1 \\ e_{12} &= -e_{21} \text{ ; short hand notation} \end{aligned}$$

This means that we can swap the order of the product of two perpendicular basis vectors but we must include a minus sign to enforce equality. The arc $e_2 e_1$ sweeps 90° CW not CCW. This is why we must reflect e_2 to make the arc CCW.

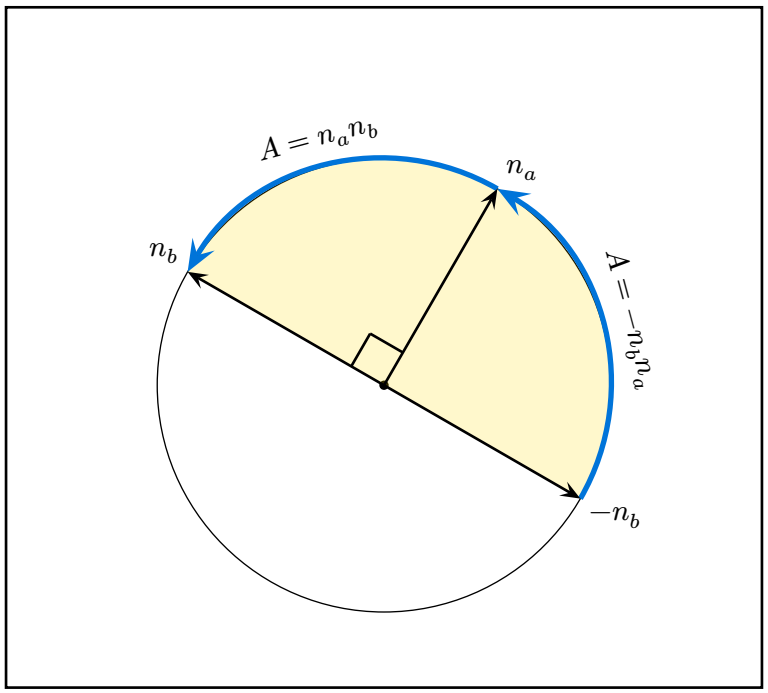

Figure 20: Equal 90° arcs

We can extend this result to any pair of perpendicular vectors. In Figure 20, both arcs are 90° CCW. Sweeping from $\vec{n}_a$ to $\vec{n}_b$ by 90° is equivalent to sweeping from $-\vec{n}_b$ to $\vec{n}_a$. Thus $n_a n_b = -n_b n_a = e_1 e_2$. This is true for any pair of unit vectors separated by 90° CCW because of the principle of equal arcs in postulate four. If $a = r_a \vec{n}_a$ and $b = r_b \vec{n}_b$, where a is perpendicular to b, we can use these facts to show that $ab = -ba$:

$$\begin{aligned}
ab &= (r_a \vec{n}_a)(r_b \vec{n}_b) \text{ ; move scalars to the left} \\
&= (r_a r_b)(\vec{n}_a \vec{n}_b) \text{ ; use } \vec{n}_a \vec{n}_b = -\vec{n}_b \vec{n}_a \\
&= (r_a r_b)(-\vec{n}_b \vec{n}_a) \text{ ; move scalar } r_a \text{ right and '-' left} \\
&= -(r_b \vec{n}_b)(r_a \vec{n}_a) \\
ab &= -ba
\end{aligned}$$

This relationship is extremely important. We have derived it without referring to the dimensions for the vectors a and b. They must be at least two dimensional but could be three or more dimensional vectors and the formula would still be true!

The two vectors will specify a unique plane that contains them both and the arc will lie in that plane. We can then shift our point of view to "look down" on the plane as if it was a sheet of paper and rotate the paper to any angle so that the vector b was pointing to the right for example. Nothing would change. Changing your point of view has no effect on the geometry of the situation. It just amounts to choosing a different set of basis vectors to measure other vectors with. This is sometimes referred to as *coordinate free* geometry. It simplifies the formulas but more importantly it makes them ***universal***, greatly extending their reach.

3.3. Zero Degree Arc

What happens when the angle between the unit vectors in an arc is zero? The staring vector is the same as the ending vector: $B = n_1 n_1$. The value of arc B cannot depend on the direction of n_1 since we are free to rotate both together without changing B (equal 0° arcs). We no longer know which plane the arc is in since the two vectors are identical. The only information we have left is the radius of the arc which is the length of the unit vector n_1. But this is a length of one unit. We will ***define*** the product of a unit vector with itself to be the number one: $n_1 n_1 = 1$ for any unit vector regardless of its direction. This leads to another vital relationship between *parallel* basis vectors:

$$\begin{aligned} e_1^2 &= e_1 e_1 \ = 1 \\ e_2^2 &= e_2 e_2 \ = 1 \\ \vec{n}_1^2 &= \vec{n}_1 \vec{n}_1 = 1 \end{aligned}$$

The simple relationships between parallel and perpendicular vectors is why it is best to choose basis vectors that are mutually perpendicular unit vectors. It greatly simplifies multiplying vectors together.

The ***contraction*** rule from the last chapter is a direct consequence of this definition. Given an arbitrary vector $\vec{a} = r_a \vec{n}_a$ where $|a| = r_a$ is the length of the vector (a scalar) and $\vec{n}_a$ is a unit vector in the direction of $\vec{a}$:

$$\begin{aligned} \vec{a}^2 &= \vec{a}\vec{a} \\ &= (r_a \vec{n}_a)(r_a \vec{n}_a) \text{ ; move scalars to the left} \\ &= r_a r_a \vec{n}_a \vec{n}_a \\ &= r_a^2 \vec{n}_a^2 \text{ ; use } \vec{n}_a^2 = 1 \\ &= r_a^2 (1) \\ &= r_a^2 \text{ ; this is a scalar} \\ \vec{a}^2 &= r_a^2 = |a|^2 \text{ ; the contraction rule} \end{aligned}$$

This rule can be extended to any pair of *parallel* vectors $a = r_a \vec{n}$ and $b = r_b \vec{n}$. We say they are parallel because they share the same direction vector $\vec{n}$.

$$\begin{aligned} ab &= (r_a \vec{n})(r_b \vec{n}) \text{ ; move scalars to the left} \\ &= (r_a r_b)(\vec{n}\vec{n}) \\ &= r_a r_b \vec{n}^2 \text{ ; use } \vec{n}^2 = 1 \text{ by definition} \\ &= r_a r_b (1) \\ &= r_a r_b \text{ ; this is a scalar} \\ &= |a||b| \end{aligned}$$

It is hard to overstate the importance of the contraction rule. It is the glue that holds geometric algebra together as we shall see. We can also show that if (and *only* if) the vectors are parallel are we free to swap their order.

$$
\begin{aligned}
ab &= (r_a\vec{n})(r_b\vec{n}) \text{ ; move scalars to the left} \\
&= (r_a r_b)(\vec{n}\vec{n}) \text{ ; move } r_a \text{ to the right} \\
&= (r_b\vec{n})(r_a\vec{n}) \\
ab &= ba
\end{aligned}
$$

There are two universal rules for vector multiplication:

$$
\begin{aligned}
ab &= -ba \text{ ; when } a \text{ is } \textbf{perpendicular} \text{ to } b \\
ab &= ba \text{ ; when } a \text{ is } \textbf{parallel} \text{ to } b
\end{aligned}
$$

When a and b are perpendicular vectors, their product is a scaled 90° arc. This is often called a ***bivector***. When a and b are parallel vectors, the their product is always a scalar.

3.4. The Right Angle Triangle

Problem: find the length of the longest side of a right triangle labeled $c = a + b$ given the lengths of the two perpendicular sides labeled a and b as shown in Figure 21. We can find the length of a vector using the contraction rule. Since the vectors a and b are perpendicular we can also use the relationship $ab = -ba$ derived above. We will also use the "sum of products" result from Section 2.11: $(a + b)(a + b) = aa + ab + ba + bb$.

$$
\begin{aligned}
c^2 = cc &= (a + b)(a + b) \text{ ; use sum of products} \\
&= aa + ab + ba + bb \text{ ; rearrange the products} \\
&= a^2 + b^2 + ab + ba \text{ ; use } ab = -ba \\
&= a^2 + b^2 + -ba + ba \text{ ; label } ba = B \\
&= a^2 + b^2 + (-B + B) \text{ ; use} - B + B = 0 \\
&= a^2 + b^2 + 0 \\
c^2 &= a^2 + b^2 \text{ ; use the contraction rule: } a^2 = |a|^2 \\
|c|^2 &= |a|^2 + |b|^2
\end{aligned}
$$

This is the ***Pythagorean Theorem***. It is perhaps the most famous theorem in all of mathematics. Euclid provides an ingenious construction for this relationship in his book but it is quite complicated. Our derivation is much simpler and works in any number of dimensions.

3.5. The Carpenter's Square

For example, the famous triangle with side lengths 3, 4 and 5 is shown in Figure 21. It is sometimes called the Carpenter's Square. References to it go back over 5000 years. If three sticks are cut to those lengths and pinned together the resulting triangle will have a 90° angle between the shorter sides.

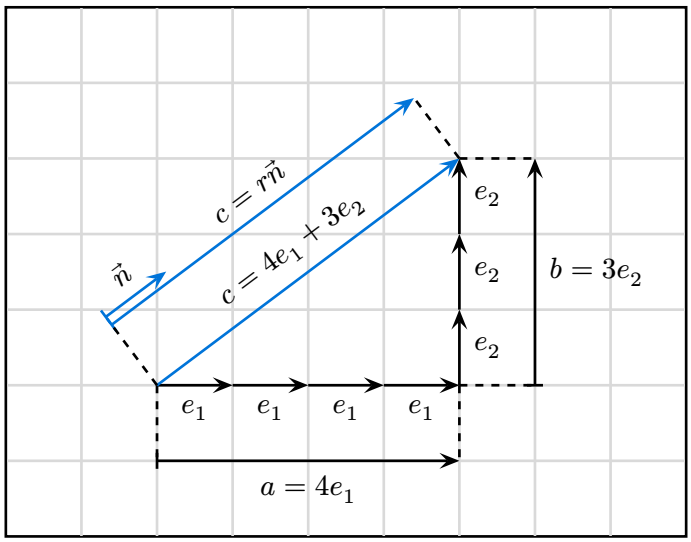

Figure 21: The Carpenter's Square

We can use the contraction rule to get the length of vector c:

$$\begin{aligned}
a &= 4e_1 \\
b &= 3e_2 \\
c &= r\vec{n} = a + b \text{ ; } a \text{ and } b \text{ are perpendicular} \\
c^2 &= a^2 + b^2 \text{ ; use the Pythagorean Theorem} \\
(r\vec{n})^2 &= (4e_1)^2 + (3e_2)^2 \\
r^2\vec{n}^2 &= 4^2e_1^2 + 3^3e_2^2 \\
r^2(1) &= 16(1) + 9(1) \\
r^2 &= 25
\end{aligned}$$

Thus $|c| = r = 5$ as expected, since $5^2 = 25$.

We have to be careful to apply the Pythagorean theorem only to vectors composed of a sum of **perpendicular** vectors. For example, in Figure 15, $a = 7e_1 + 2e_2$, $b = -2e_1 + 5e_2$ and $c = a + b$. We can't use $c^2 = a^2 + b^2$ because the vectors a and b are **not** perpendicular. But if we first add a and b to get $c = 5e_1 + 7e_2$, we can now use Pythagoras to get the length of c since the basis vectors *are* perpendicular.

$$\begin{aligned}
c &= 5e_1 + 7e_2 \\
c^2 &= 5^2 + 7^2 = 25 + 49 = 74 \\
|c| &= \sqrt{74} = 8.6
\end{aligned}$$

Check that $a^2 + b^2 = 53 + 29 = 82 \neq 74$. For the three dimensional example in Figure 17, $c = e_1 + e_2 + e_3$. The three basis vectors are mutually perpendicular so we can use Pythagoras:

$$\begin{aligned}
c^2 &= e_1^2 + e_2^2 + e_3^2 \\
&= 1 + 1 + 1 = 3 \\
|c| &= \sqrt{3} = 1.73
\end{aligned}$$

This is another reason why it's very handy to use mutually perpendicular basis vectors to represent arbitrary vectors.

3.6. The Written Form of a Unit Vector

Since arcs are composed of unit vectors which can point in any direction, we need a way to write them down. If we already have a vector we can convert it to a unit vector by simply dividing by its length, which we can get using the contraction rule. For example:

$$\begin{aligned}
c &= |c|\vec{n} = 3e_1 + 4e_2 \text{ ; the carpenter's square} \\
c^2 &= 3^2 + 4^2 \text{ ; using Pythagoras} \\
&= 9 + 16 = 25 \\
|c| &= \sqrt{25} = 5 \\
c &= 5\vec{n} = 3e_1 + 4e_2 \text{ ; divide both sides by 5 to isolate } \vec{n} \\
\vec{n} &= \left(\frac{1}{5}\right)(3e_1 + 4e_2) \\
\vec{n} &= \frac{3}{5}e_1 + \frac{4}{5}e_2
\end{aligned}$$

The unit vector is composed of a "bit" of the horizontal vector $\frac{3}{5}e_1$ and a "bit" of the vertical vector $\frac{4}{5}e_2$. The result is a vector directed "between" the horizontal and vertical directions. This is generally the case when adding geometric elements of the same type. For example we will see later that the sum of two lines will be a line between them passing thru their intersection point and the sum of two points will likewise be a point between them along the line connecting the points.

If we are given the angle of the unit vector relative to a reference direction we can use a calculator to compute two scalars called the "cos" and "sin" values for that angle. The angle is usually labeled with a lowercase t and the basis vector e_1 is usually used as the reference direction. For example in Figure 22, $\vec{n}_1$ is a unit vector 60° CCW from e_1:

$$\begin{aligned}
t_A &= 60° \text{ ; angle from } e_1 \text{ to } \vec{n}_1 \\
c &= \cos(t_A) = \cos(60°) = 0.5 \text{ ; use a calculator} \\
s &= \sin(t_A) = \sin(60°) = 0.87 \text{ ; use a calculator} \\
\vec{n}_1 &= ce_1 + se_2 = 0.5e_1 + 0.87e_2
\end{aligned}$$

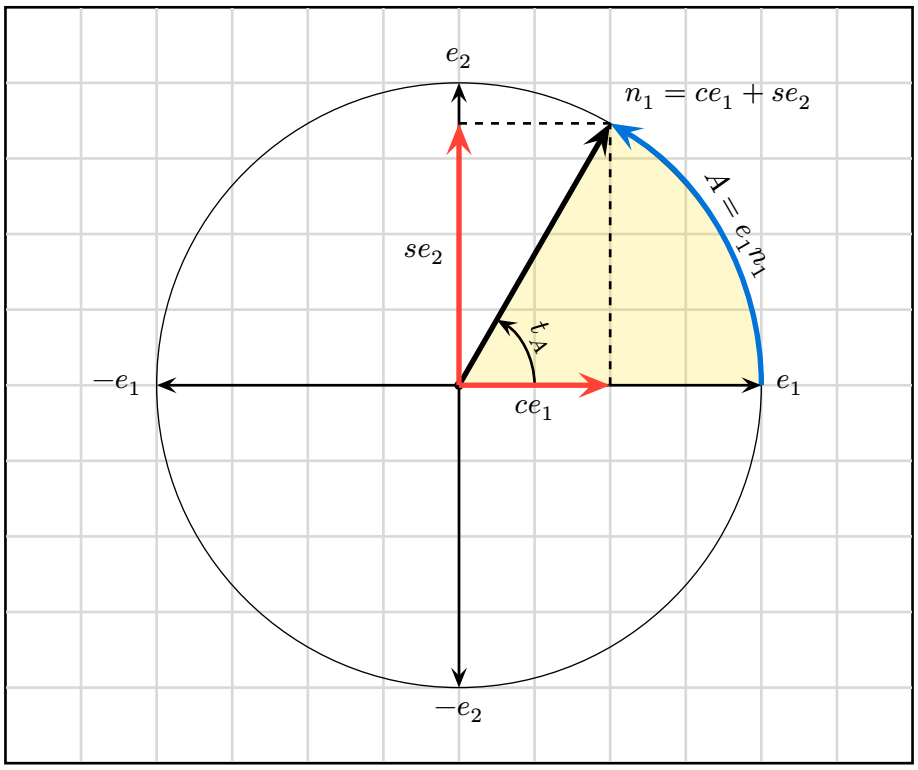

Figure 22: Unit Vector

We can think of the "cos" scalar as the length of the projection or shadow of the unit vector n_1 in the e_1 direction. Similarly the "sin" scalar is the length of the projection of n_1 in the e_2 direction. Euclid provides a way of constructing these projections with a straight edge and compass. The "cos" and "sin" projections ensure that the unit vector squares to one for any angle.

$$\begin{aligned} n &= ce_1 + se_2 \\ n^2 &= (ce_1)^2 + (se_2)^2 \text{ ; use Pythagoras} \\ &= c^2 e_1^2 + s^2 e_2^2 \\ &= c^2(1) + s^2(1) \\ n^2 = 1 &= c^2 + s^2 \end{aligned}$$

Mathematicians have discovered various clever and efficient recipes for calculating the "cos" and "sin" projection values. Before we had electronic or mechanical calculators we used human "calculators" to compute tables of "cos" and "sin" values for angles from 0° to 90° accurate to 4 or 5 decimal places. The values for the other angles could be derived from the first quadrant values with some sign changes. These tables were compiled into very valuable books that were used for astronomy, navigation and surveying among many other uses. The values were computed "by hand" in an extremely laborious and error prone process. We are very lucky to live in a time when we all carry around super computers in our pockets!

Unit vectors will usually be written as $\vec{n}_1 = ce_1 + se_2$ where $c = \cos(t)$, $s = \sin(t)$ and t is the angle from e_1 to $\vec{n}_1$. We can now write down the value of an arc in terms of our basis vectors:

$$
\begin{aligned}
A &= e_1 n_1 \\
&= e_1(ce_1 + se_2) \text{ ; distribute } e_1 \text{ and move scalars left} \\
&= ce_1 e_1 + se_1 e_2 \\
&= ce_1^2 + se_1 e_2 \\
&= c(1) + se_1 e_2 \\
A &= c + se_{12}
\end{aligned}
$$

This shows two ways of writing down an arc:
1. The product of two unit vectors, $A = n_1 n_2$
2. The sum of a scalar and a bivector: $A = c + se_{12}$

Both forms are very useful as we shall see. The arc A is a sum of a bit of the zero degree arc ($ce_1^2 = c$) and a bit of the ninety degree arc (se_{12}) : $A = c(1) + s(e_{12})$. So the angle of the arc will be *between* 0° and 90° depending on the values of c and s (for positive values of c and s). For example if $c = s$, we will have equal contributions so the angle t_A will be half way between 0° and 90° giving $t_A = 45°$.

Any pair of perpendicular reference unit vectors can be used in place of the basis vectors. For example, if we use f_1 and f_2 as our perpendicular reference vectors, we can write a unit vector n_1 in terms of these reference vectors as follows: $n_1 = cf_1 + sf_2$ where $c = \cos(t)$ and t is the angle from f_1 to n_1. Note that $f_1 f_2 = e_1 e_2$ since they are both 90° arcs in the same plane where the f vectors are rotated with respect to the basis vectors by any angle we like. The result is exactly the same arc as above:

$$
\begin{aligned}
A &= f_1 n_1 \\
&= f_1(cf_1 + sf_2) \\
&= cf_1 f_1 + sf_1 f_2 \text{ ; use } f_1 f_2 = e_1 e_2 \\
&= cf_1^2 + se_1 e_2 \\
&= c(1) + se_1 e_2 \\
A &= c + se_{12} \text{ ; as before}
\end{aligned}
$$

This is a verification of the equal arcs principle and is very useful since it allows one to pick a convenient "point of view" when writing down unit vectors and arcs.

3.7. Rotations

The product of a vector and a scalar is a vector. The product of a vector and another vector is an arc. What is the product of a vector and an arc (or equivalently the product of three vectors)? It's easy to show that the result is a new vector. In Figure 23 we have two equal arcs: $A = n_1 n_2 = n_3 n_4$.

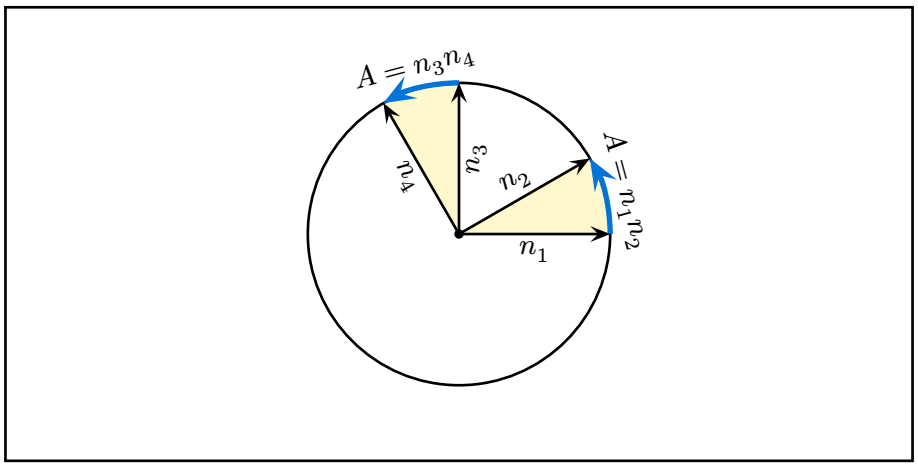

Figure 23: The product of a vector and an arc

If we multiply n_3 by A we get n_4:

$$\begin{aligned} n_3 A &= n_3(n_1 n_2) \text{ ; use } n_1 n_2 = n_3 n_4 \\ &= n_3(n_3 n_4) \\ &= n_3^2 n_4 \text{ ; use } n_3^2 = 1 \\ n_3 A &= n_4 \end{aligned}$$

This shows that multiplying a vector n_3 by an arc A will be another vector n_4 which will be equal to n_3 rotated by the angle of the arc. The starting vector of A doesn't matter. Neither does the direction of n_3. This is a universal way of rotating a vector in two dimensions (2D). What happens if we swap the order of the vector and the arc?

$$\begin{aligned} A n_4 &= (n_1 n_2) n_4 \\ &= (n_3 n_4) n_4 \\ &= n_3 n_4^2 \\ A n_4 &= n_3 \end{aligned}$$

Left-multiplying a vector by arc A will rotate it in the opposite orientation of the arc: n_4 is rotated CW instead of CCW to n_3.

For example, e_1 points right. If we rotate it 90° CCW with the arc $A = e_{12}$ then it will point up in the e_2 direction. Let's try it:

$$\begin{aligned} e_1 A &= e_1(e_1 e_2) \\ &= e_1^2 e_2 \text{ ; use } e_1^2 = 1 \\ e_1 A &= e_2 \text{ ; as expected} \end{aligned}$$

If we rotate e_2 by 90° CCW it will point left in the $-e_1$ direction:

$$
\begin{aligned}
e_2 A &= e_2(e_1 e_2) \text{ ; use equal arcs } e_1 e_2 = -e_2 e_1 \\
e_2 A &= e_2(-e_2 e_1) \\
&= -e_2^2 e_1 \text{ ; use } e_2^2 = 1 \\
e_2 A &= -e_1 \text{ ; as expected}
\end{aligned}
$$

As an exercise try rotating the Carpenter's Square, $c_1 = 4e_1 + 3e_2$ by 90° CCW using $A = e_{12}$. You should get $c_2 = c_1 A = -3e_1 + 4e_2$. You can also verify that c_1 is perpendicular to c_2 by simply multiplying them together. If they are perpendicular then the result should be a scaled 90° arc. (Answer: $c_1 c_2 = 25e_{12}$).

Of course we can use this idea to rotate any vector by any arc as long as the vector lies in the same plane as the arc. Later we will show how to extend rotations to any vector. For example if you want to rotate a vector by 60°, construct the arc $A = c + se_{12}$ where $c = \cos(60°) = 0.5$ and $s = \sin(60°) = 0.87$ using a calculator. Then multiply the vector by A.

3.8. The Product of Two Arcs

We have shown that multiplying three vectors results in a single vector. If we multiply four vectors or two arcs we should expect to get an arc since three of the vectors becomes one so this is the same as multiplying two vectors which is an arc. It's easy to show that this is indeed the case as shown in Figure 24.

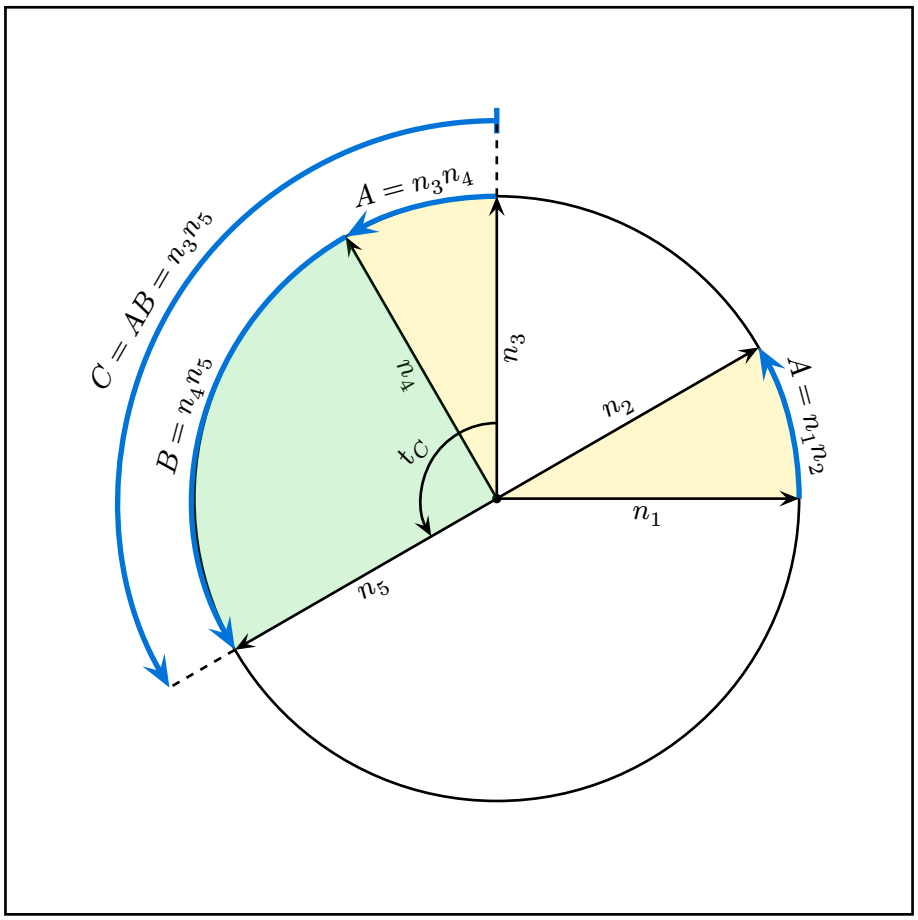

Figure 24: The Product of Two Arcs

$$
\begin{aligned}
C &= AB \\
&= (n_1 n_2)(n_4 n_5) \text{ ; use } n_1 n_2 = n_3 n_4 \\
&= (n_3 n_4)(n_4 n_5) \\
&= n_3 n_4^2 n_5 \text{ ; use } n_4^2 = 1 \\
C &= n_3 n_5 \\
t_C &= t_A + t_B
\end{aligned}
$$

Notice that the resulting arc C has an angle equal to the ***sum*** of the angles of A and B. The product of two arcs is always an arc, even if the two arcs are "separated" as shown in Figure 24.

An interesting example of the product of two arcs is the product of e_{12} with itself:

$$
\begin{aligned}
A &= e_{12} \\
A^2 &= AA \\
&= e_{12}e_{12} \\
&= (e_1e_2)e_1e_2 \text{ ; use } e_1e_2 = -e_2e_1 \\
&= (-e_2e_1)e_1e_2 \\
&= -e_2e_1^2e_2 \\
&= -e_2e_2 \\
&= -e_2^2 \\
e_{12}^2 &= -1
\end{aligned}
$$

At first glance this result may seem odd since the square of a number is always positive. As shown above, when we multiply arcs we add their angles. The product of two 90° arcs must be a $90° + 90° = 180°$ arc. For example the arc B from e_1 to $-e_1$ is also a 180° arc and $B = e_1(-e_1) = -(e_1^2) = -1$. This is also consistent with the "sum-form" of an arc $B = c + se_{12}$. If the angle $t_B = 180°$ then $c = \cos(180°) = -1$ and $s = \sin(180°) = 0$, thus $B = -1 + 0e_{12} = -1$ as expected. So we can think of -1 as a 180° arc or **rotor** . The result is the same as a point reflection in 2D.

Note that all the arcs are in the same plane since in Euclidian plane geometry there is only one plane. Later we will extend these ideas to 3 or more dimensions where there are an infinite number of planes.

3.9. Multivectors

We have shown that if we multiply an odd number of vectors together the result is a single vector and the product of an even number of vectors results in a single arc. It seems that all geometric elements in 2D can be represented as a sum of a vector and an arc:

$$
\begin{aligned}
\vec{a} &= a_1e_1 + a_2e_2 \\
A &= c + se_{12} \\
\vec{a} + A &= a_1e_1 + a_2e_2 + c + se_{12} \\
&= c(1) + a_1(e_1) + a_2(e_2) + s(e_{12})
\end{aligned}
$$

We have four scalars times four basis elements $\{1, e_1, e_2, e_{12}\}$. This is similar to a vector except it can represent more elements than just vectors. We call such an arbitrary geometric element, a ***multivector***. The basis elements can be formed by multiplying basis vectors together in a systematic way. For example the basis element $1 = e_1e_1$ and the basis element $e_{12} = e_1e_2$. The basis vectors will determine a unique set of basis elements for the space. Every geometric element will then have a unique representation as a multivector by scaling and adding the basis elements. Some examples:

$$
\begin{aligned}
5(1) + 0(e_1) + 0(e_2) + 0(e_{12}) &= 5 \text{ ; a scalar} \\
0(1) + 3(e_1) + 4(e_2) + 0(e_{12}) &= 3e_1 + 4e_2 \text{ ; a vector} \\
0(1) + 0(e_1) + 0(e_2) + 7(e_{12}) &= 7e_{12} \text{ ; a bivector} \\
-0.5(1) + 0(e_1) + 0(e_2) + 0.87(e_{12}) &= -0.5 + 0.87e_{12} \text{ ; an arc}
\end{aligned}
$$

The rules of addition and multiplication we have introduced will guarantee that the sum of any two multivectors will be a multivector and the product of any two multivectors will be a multivector. This forms a closed system where all possible geometric elements that can be formed using our basis elements will each have a unique representation. The basis elements can be categorized by *grade*. The grade of an element is the number of perpendicular vectors that are multiplied together to form it. For example the basis element e_{12} has a grade of 2 since it's the product of 2 perpendicular vectors. The grade of a scalar is defined as 0 and the grade of a vector is defined as 1.

We can represent more complex geometries by simply adding more basis vectors to the system. For example, to represent a 3D space we can add another basis vector called e_3 and generate the new basis elements for the 3D space. We will inherit the four previous basis elements $\{1,e_1,e_2,e_{12}\}$ and generate four more by multiplying each of them by e_3:

$$\{1, e_1, e_2, e_{12}\} \times e_3$$
$$\{(1)e_3, (e_1)e_3, (e_2)e_3, (e_{12})e_3\}$$
$$\{e_3, e_{13}, e_{23}, e_{123}\}$$

The full set of basis elements for a 3D multivector will be:

$$\{1, e_1, e_2, e_3, e_{12}, e_{13}, e_{23}, e_{123}\}$$

We have eight basis elements that can be used to represent the much richer geometry of 3D space. There are 3 grade 1 basis vectors $\{e_1, e_2, e_3\}$, 3 grade 2 basis bivectors $\{e_{12}, e_{13}, e_{23}\}$ and 1 grade 3 basis trivector e_{123}. The three basis vectors can be combined to form arbitrary 3D vectors. The three bivector elements can be combined to represent arbitrary 3D planes. Arcs are created by multiplying two 3D unit vectors together which can then be used as rotors to do complex rotations in 3D space. Fortunately we don't have to learn any new rules. The power of the geometric product will do most of the work for us. And we don't have to stop at a paltry three dimensions…

3.10. Signatures

We defined the square of a unit vector as 1 since that described the radius of a unit circle. It turns out that we are actually free to define the square of our basis vectors to be any number we like, as long as the choice makes sense for the geometry we are trying to represent. The most convenient choices turn out to be one, negative one or zero $\{1, -1, 0\}$. These numbers are sometimes called the signature of the basis element. Spatial direction vectors like e_1,e_2,e_3 usually have a signature=1. We will see later that a basis vector that squares to zero (signature = 0) will be extremely useful for representing parallel lines. It is also central to the branch of mathematics called ***calculus***. A basis vector with a signature = −1 can be used to capture the physical effects of Einstein's relativity theory and Maxwell's theory of electromagnetism.

3.11. Summary

Starting with a minimum set of freely chosen basis vectors as our building blocks we can create new geometric elements by adding and multiplying them together. Geometric Algebra gives us a universal language to write down any geometry and manipulate the geometric elements therein by the same set of simple universal rules.

This language can be used to model 3D objects in computer graphics and games. It can model the 4D word of Einstein's theory of relativity and Maxwell's 6D theory of electromagnetics. It can model the exotic spinors and twistors of quantum theory as well as the quantum bits in a quantum computer. We can even model the curved geometry of manifolds like Einstein's theory of gravity using Geometric Calculus. Geometric Algebra is an immensely powerful tool.

The number of basis vectors we need for any particular problem situation should be chosen to be as small as possible (but no smaller). So far we have chosen the unit vectors e1 and e2 to carry information about spatial directions on a flat plane. The information is carried by the numbers we use to scale the basis vectors. The scalars will determine the direction and length of the resulting vectors.

We have covered everything we need to extend geometric algebra to any number of dimensions but it turns out that we still need one more basis vector to complete Euclid's plane geometry. This is because we have no written form for two parallel lines that are separated by a particular distance as required by Euclid's fifth postulate. This highlights the importance of that troublesome postulate. It took over two thousand years to find the smallest geometric algebra capable of fully representing a *Euclidean* 2D space! A very clever and powerful way to do that has recently been discovered and will be explored next. Fortunately no new rules will be required, just a new basis vector.